

Small Language Model Optimization on WikiText 2

DASE7506 MP1 30 September 2026

Wang Ziyue 3036709042

A width-288, depth-4 causal Transformer with RMSNorm, RoPE, SwiGLU, MHA, and 0.15 residual dropout is the best validation-selected model in the supplied runs. Its minimum validation score is 1.50726 BPB. The recorded full-test score is 1.53231 BPB for the matching final-step checkpoint, compared with 2.10191 BPB for the baseline. In a paired CPU rerun, scoring took 2.75 times the baseline time; peak process RSS was 2.305 GiB versus 2.275 GiB for baseline. Both are within the measured 5x time and 4 GiB RAM limits. The checkpoint is 19.71 MiB.

Key Results

Measure	Baseline	Final recorded checkpoint
Full-test BPB	2.10191	1.53231
Validation BPB	2.07181	1.50788 final; 1.50726 best
Parameters	1,088,256	4,573,728
CPU test time	5.42 s	14.91 s (2.75x)
Checkpoint file	4.17 MiB	19.71 MiB
CPU peak RSS	2.275 GiB	2.305 GiB

Full test BPB

Fixed WikiText-2 protocol; lower is better

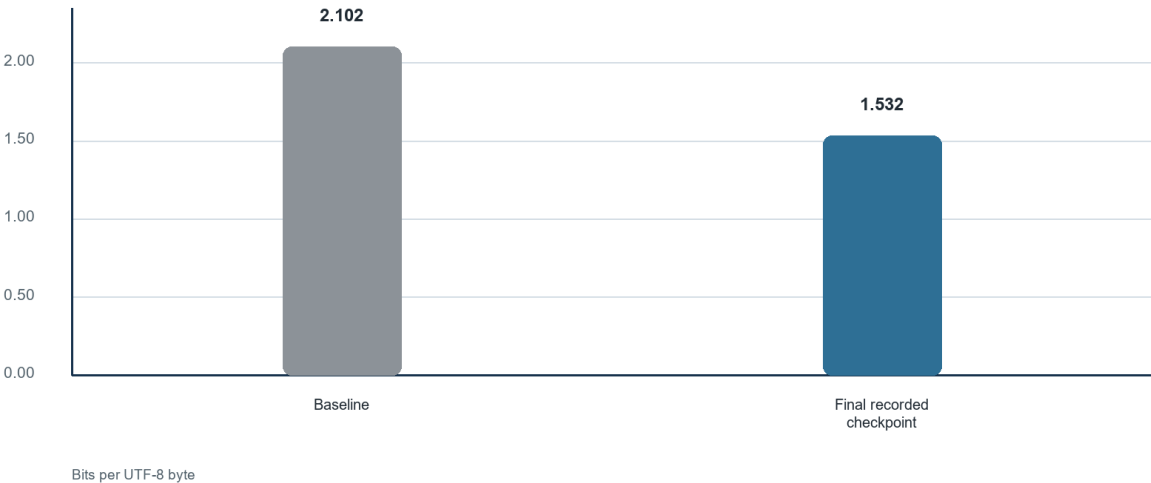


Figure 1. Full-test BPB for the supplied baseline and final recorded checkpoint.

The final checkpoint improves test BPB by 0.56960 bits per byte (27.1% relative to baseline). The validation-selected step-18,900 checkpoint reached 1.50726 BPB; the saved test JSON instead matches checkpoint.pt at step 19,200, whose validation BPB is 1.50788. These are distinct checkpoints and are labeled separately throughout this report. Paired CPU rerun: baseline 5.42 s / 2.275 GiB peak RSS; final model 14.91 s / 2.305 GiB. The peak RSS was obtained from `Linux resource.getrusage(RUSAGE_CHILDREN).ru_maxrss`, converted from KiB to GiB.

Task and Evaluation

The benchmark is WikiText-2 raw text with the supplied train-fitted BPE-2048 tokenizer. The protocol fixes context length at 256 and scores independent causal windows. The scorer sums the negative log-base-2 probability assigned to each next token and divides by the split's raw UTF-8 byte count. Validation has 376,599 targets and 1,148,007 bytes; test has 428,405 targets and 1,292,013 bytes. Lower BPB is better.

Development and checkpoint selection use validation. Training uses only the supplied training text. The tokenizer, context length, evaluator, and benchmark windows remain fixed; all models predict causally and use no pretrained weights or external training text.

Model Changes

Component	Baseline	Best validation configuration
Shape	Width 128, depth 4, 4 heads	Width 288, depth 4, 4 heads
Position	Learned absolute position embeddings	RoPE
Normalization	Pre-LayerNorm	Pre-RMSNorm
Attention	Causal MHA	Causal MHA; 4 query and 4 KV heads
Feed-forward	4x GELU MLP	SwiGLU, hidden multiplier 8/3
Linear biases	Present in block linears	Removed
Embedding tie	Token/output weights tied	Retained
Dropout	None	Residual 0.15; embedding 0
Parameters	1,088,256	4,573,728

The best model is MHA, not GQA: its query-head and KV-head counts are both four. RoPE, RMSNorm, SwiGLU, and bias removal were introduced together in the combined architecture; the saved run set does not contain a complete one-at-a-time ablation of each component. Their individual effects therefore cannot be isolated from the cumulative baseline-to-model gain.

Final Training Recipe

Setting	Value
Optimizer and precision	AdamW; BF16 training, FP32 scoring
Peak learning rate and warmup	0.0015; 100 steps
Cosine schedule floor	0.01 of peak rate
Weight decay and dropout	0.4; residual dropout 0.15
Batch, updates, seed	32; 19,200; seed 71
Processed target tokens	157,286,400
Training device	NVIDIA GeForce RTX 5090

Experimental Design and Training Budget

Experiments moved from small width and learning-rate screens to regularization, attention/FFN variants, training-horizon comparisons, and matched-seed architecture comparisons. Most later runs use seed 71, width 288, batch 32, maximum learning rate 0.0015, 100-step warmup, cosine floor 0.01, and validation checks every 300 steps. Shorter exploratory runs are not treated as directly comparable to the 19,200-step finalists unless their settings match.

Equal Target Budget Comparison

Run at 1,200 updates	Processed targets	Validation BPB
Baseline GPT, seed 17	9,830,400	2.07181
Width 288 combined model, seed 17	9,830,400	1.74232

Both runs process $1,200 \times 32 \times 256 = 9,830,400$ next-token targets. This satisfies a target-budget comparison and shows that the developed model learns useful structure faster under the tested recipes. It is not a pure architecture ablation: the model, optimizer settings, and learning-rate schedule differ, and the combined-model score is read at step 1,200 from a run whose cosine horizon is 14,400 steps.

Longer Training and Learning Rate Horizon

Seed-71 MHA4 run	Best validation BPB	Best step	Targets
14,400-step cosine plan	1.53343	14,400	117.96M
19,200-step cosine plan	1.50726	18,900	157.29M

Validation learning curves

Width 288, depth 4, seed 71, WD 0.4, dropout 0.15; cosine horizon differs

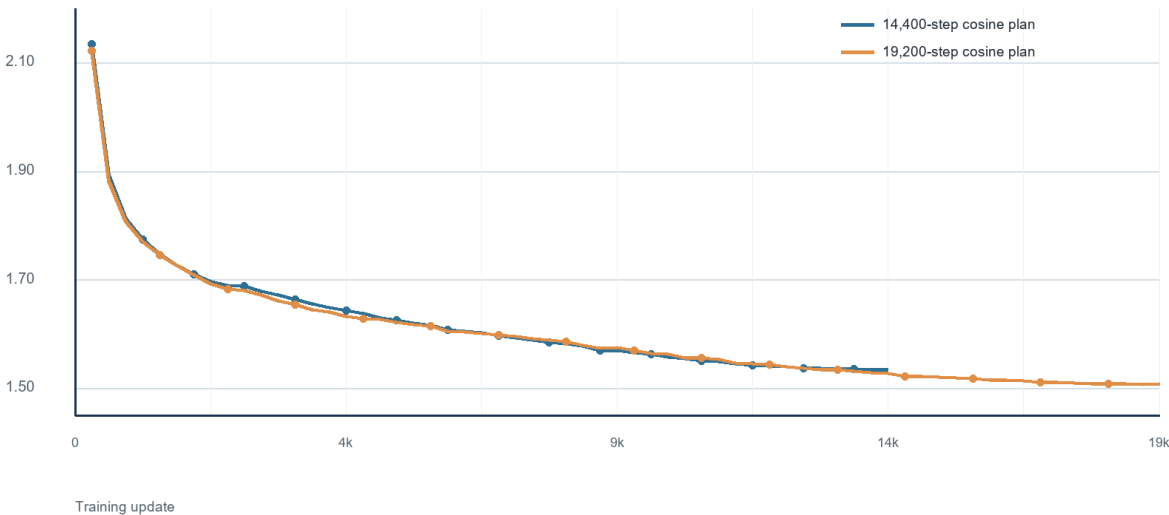


Figure 2. Validation curves. Extending the total plan also stretches the cosine schedule, so the gain combines more updates with a different learning-rate trajectory.

Validation BPB continued to improve late in the longer run, with its minimum at step 18,900 and a small increase at step 19,200. This is evidence that the shorter recipe had not reached the same validation level, but it does not identify whether extra updates or slower decay caused the gain. The best-versus-final difference is small and does not show a sustained overfitting rebound.

Regularization and Optimization Results

Residual Dropout

Residual dropout	Train BPB	Validation BPB	Validation minus train
0.10	1.14226	1.52165	0.37940
0.15	1.18587	1.51270	0.32683
0.20	1.23307	1.53338	0.30031

Residual dropout ablation

Seed 17, 14,400 steps, WD 0.4; same width and attention configuration

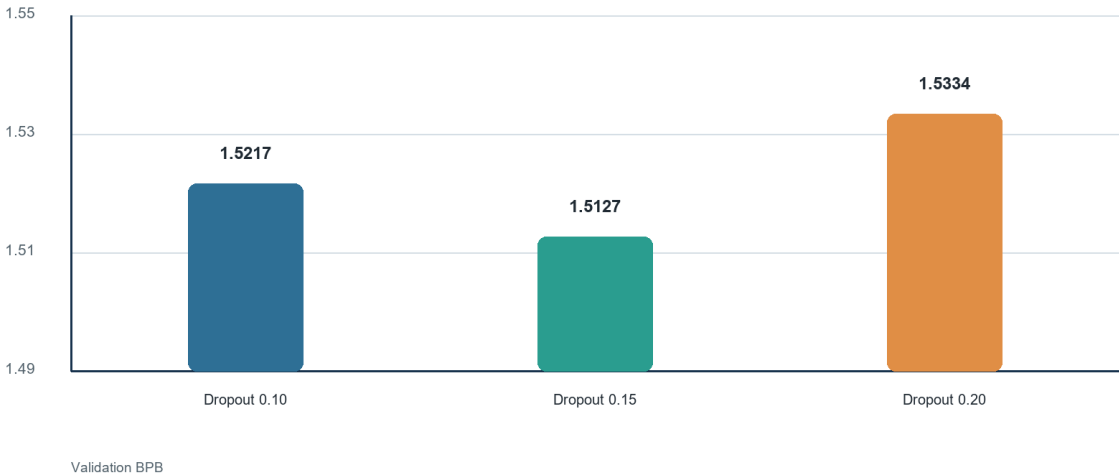


Figure 3. Validation BPB by residual-dropout probability. All runs use seed 17, WD 0.4, and 14,400 updates.

At this recipe, dropout 0.15 gives the lowest validation BPB. Dropout 0.20 reduces the train-validation gap further but worsens validation, showing that a smaller gap alone is not a selection objective. The 0.15 setting sacrifices some train fit relative to 0.10 while improving validation by about 0.009 BPB.

Weight Decay and Cosine Floor

WD, seed 71, 19.2k steps	Best validation BPB
0.004	1.54353
0.02	1.54953
0.20	1.51969
0.30	1.52501
0.40	1.50726

Weight decay ablation

Seed 71, 19,200 steps, width 288, dropout 0.15; cosine floor 0.01

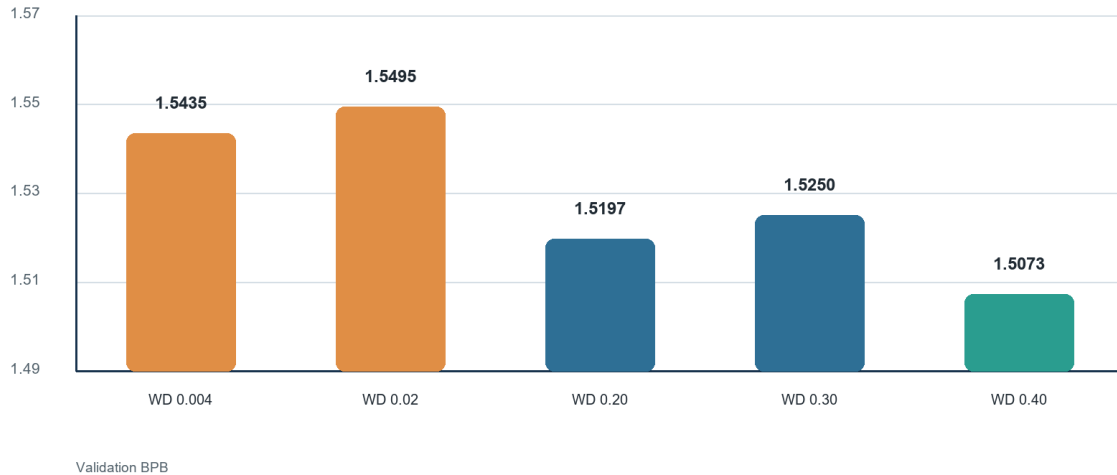


Figure 4. Weight-decay sweep under the matched 19,200-step seed-71 recipe.

The 0.4 setting is best among these longer-run points; the response is not monotonic, and seed-17 14,400-step runs put WD 0.3 and 0.4 nearly tied while WD 0.5 performed worse. A second seed also showed about 0.021 BPB variation at 14,400 steps, larger than the WD 0.3-versus-0.4 difference. This limits the precision of fine WD rankings.

The cosine floor matters: under the 19,200-step seed-71 recipe, a minimum-LR ratio of 0.01 reached 1.50726 BPB, compared with 1.52356 at ratio 0.05. A separate exploratory peak-LR/WD sweep at 19,600 steps found validation scores from 1.57092 to 1.60622 for peak rates 0.0003 to 0.0008; those runs used dropout 0.10 and are not a matched comparison with the final 0.0015 recipe.

Attention and Feed Forward Ablations

Attention alternatives were screened at width 288 and depth 4 with seed 17, WD 0.4, residual dropout 0.10, and a 9,600-step schedule. MHA with four query heads and four KV heads was strongest in this screen.

Attention variant	Heads / KV heads	Parameters	Validation BPB
MHA	4 / 4	4.57M	1.52349
MHA	8 / 8	4.57M	1.54182
MHA	2 / 2	4.57M	1.56284
GQA	4 / 2	4.24M	1.55651
GQA	8 / 4	4.24M	1.56045

Attention configuration ablation

Seed 17, 9,600 steps, width 288, WD 0.4, dropout 0.10

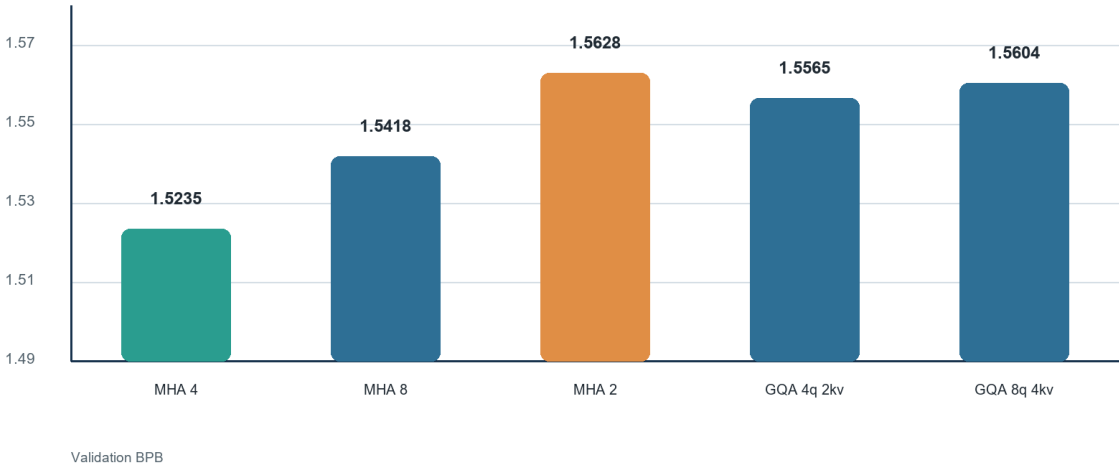


Figure 5. Head-count and KV-sharing screen; all points use the same 9,600-update recipe.

The tested GQA configurations reduced attention projection parameters but did not improve validation BPB. At this model scale and budget, the 4-head MHA variant outperformed both 8-head and 2-head alternatives. These results do not imply GQA is generally inferior; they only describe the tested configurations and training schedule.

Feed Forward Width

Variant	Multiplier	Parameters	Validation BPB
Default SwiGLU	8/3	4.57M	1.52165
Wider SwiGLU	3.0	4.91M	1.53982

The multiplier-3 FFN used more parameters without improving validation in the 14,400-step screen. No separate one-factor ablations were saved for RMSNorm, RoPE, or bias removal; these remain part of the cumulative architecture change rather than individually established causes of the gain.

Model Capacity and Alternative Structures

Configuration	Parameters	Train BPB	Best validation	Gap
width256 depth4	3,688,704	1.21628	1.55350 @ 18900	0.337
width288 depth4	4,573,728	1.15475	1.50726 @ 18900	0.353
width320 depth5	6,815,680	1.05527	1.50877 @ 18600	0.454
hybrid attention conv	4,078,944	1.16039	1.50944 @ 18900	0.349
shared 2 of 4	2,581,920	—	1.56513 @ 18900	Not measured

Model scale and validation performance
Matched seed 71, 19,200 updates, WD 0.4, residual dropout 0.15; lower BPB is better

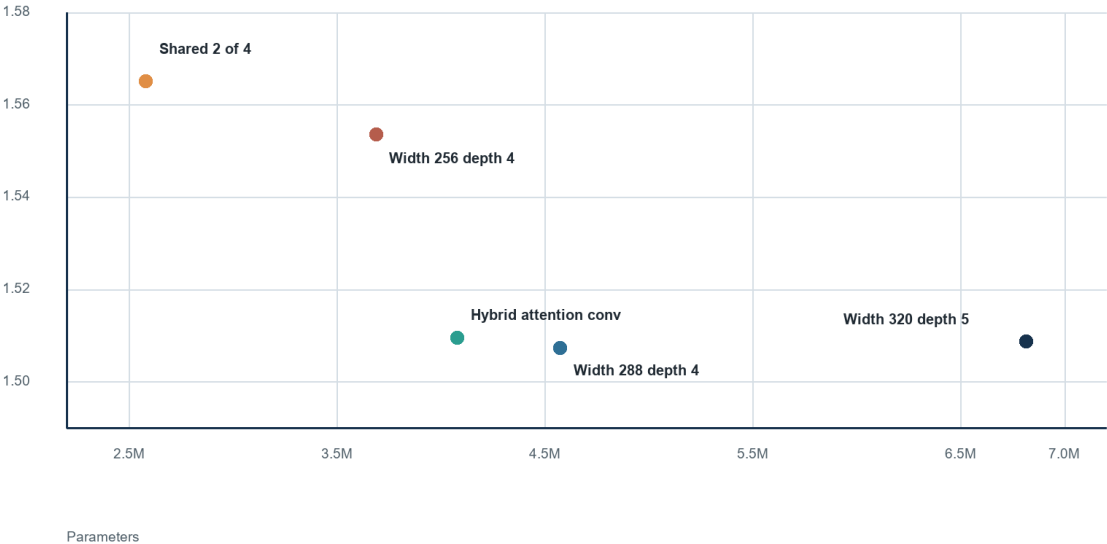


Figure 6. Capacity and alternate-structure screen. Lower validation BPB is better; all runs share seed, budget, and training recipe. Width 256 underfits relative to width 288: both train and validation BPB worsen. Width 320/depth 5 improves train BPB to 1.05527 but leaves validation slightly worse than width 288, so its additional fit does not transfer to held-out text in this run. The mixed attention-convolution model is within 0.0022 BPB of the full-attention model with about 11% fewer parameters, but it does not beat it. Sharing two blocks across four execution stages lowers parameters to 2.58M and validation reaches 1.56513, materially worse than the unshared width-288 model.

These results support width 288/depth 4 as the best observed capacity trade-off. They do not show a validation curve that rises persistently with training; rather, the larger model fits training more strongly without a validation gain. The train-validation gap is a diagnostic, not a metric to minimize independently of validation BPB.

Final Evaluation and Reproducibility

Constraint or artifact	Recorded result	Assessment
Test BPB	1.532311; 428,405 targets	Measured on final-step checkpoint
CPU scoring time	14.91 s vs 5.42 s baseline (2.75x)	Within 5x limit
Checkpoint asset	20,667,253 bytes (19.71 MiB)	Within 64 MiB limit
CPU peak process RSS	Baseline 2.275 GiB; final 2.305 GiB	Both within 4 GiB
Final checkpoint SHA-256	40d5312e33ed251a3d6faabc4f3067f9faf62325bdf0bcb754bc dd9455c68918	Matches test JSON

The 1.532311 test result is tied by SHA-256 to checkpoint.pt from step 19,200. The minimum-validation checkpoint is best_checkpoint.pt from step 18,900 with a different SHA-256 and was not the checkpoint scored in that test JSON. The step-19,200 validation score is 1.50788; the step-18,900 minimum is 1.50726. The evaluator JSON's peak_allocated_gb=0.0 is a CUDA-only field and does not measure CPU RAM. Separate Linux ru_maxrss measurements report 2.275 GiB for baseline and 2.305 GiB for the final model, both below 4 GiB.

Limitations and Conclusions

- The strongest single-factor evidence is for the residual-dropout, weight-decay, cosine-floor, and attention-head screens. Component-wise ablations of RMSNorm, RoPE, SwiGLU, and bias removal were not completed.
- Seed variation at 14,400 steps was larger than the WD 0.3-versus-0.4 difference, so fine hyperparameter rankings need replication.
- No data augmentation, ensemble, VAE encoder, or external data was used. The test set and tokenizer/context were not changed for development.
- The final test checkpoint meets the measured CPU-time (2.75x baseline), peak process RSS (2.305 GiB), and asset-size limits.

The best supported configuration is width 288/depth 4 MHA with residual dropout 0.15, WD 0.4, and a 19,200-step cosine plan. Longer training improved validation relative to 14,400 steps, while increasing model capacity or replacing half the attention layers did not improve validation. The paired rerun confirms that both peak process RSS and CPU time remain within the stated limits; the submitted checkpoint bundle should match the scored step-19,200 checkpoint.

AI assistance supported implementation iteration, experiment planning, and report drafting. All numeric results in this report are read from the supplied run artifacts.